THE TAME SYMBOL ON A ONE-DIMENSIONAL LOCAL RING IS THE NORM OF THE TAME SYMBOLS OF ITS FINITE LOCAL EXTENSIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The two same-chapter references are located at their \label lines inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Preparation of the local extensions so that both tame symbols become computable. Same chapter, Lemma labelled lemma-prepare-tame-symbol (source line 642, Stacks tag 0EAG, already counted by the card chow-prepare-tame-symbol); invoked at proof lines 967-968. Supplies the extensions B_j contained in C_j in which a and b admit local factorisations of the displayed shape; the computation of the two norms on top of it is retained in the counted span.
- Gluing finitely many finite local extensions into one finite extension. Same chapter, Lemma labelled lemma-glue-at-max (source line 526, Stacks tag 0EAE, counted earlier in this night's work); invoked at proof lines 969-970. Supplies the single finite extension B contained in C with C_j isomorphic to C_{m_j}, which is what makes the product over the maximal ideals of C meaningful.
- Transitivity of the norm in a tower of field extensions. Fields, Lemma labelled fields-lemma-trace-and-norm-tower (canonical Stacks tag 0BIJ); invoked at proof lines 994-996. Supplies the passage from the double product over (j, k) to the product over j of norms for kappa(m_{j,k})/kappa(m_j), which is the last step.

Source reference. Original lemma 0EAG: sources/stacks-chow/pinned-original-source.tex, source lines 642–642. Same-chapter pointer invoked at proof lines 967-968 to choose the extensions in which a and b admit local factorisations; the norm computation built on it is retained verbatim in the counted span.

Source reference. Original lemma 0EAE: sources/stacks-chow/pinned-original-source.tex, source lines 526–526. Same-chapter pointer invoked at proof lines 969-970 to glue

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the prepared local extensions into one finite extension; the regrouping of the two products is retained in the counted span.

Source reference. Original lemma 0BIJ: sources/stacks-chow/tags.txt, source lines 14492–14492. Cross-chapter standard prerequisite invoked once at proof lines 994-996 for transitivity of the norm; the two displays it compares are retained in the counted span.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $A \subset B$ be a finite ring extension with B/A annihilated by a power of $\mathfrak{m}$ and $\mathfrak{m}$ not an associated prime of B. For $a, b \in A$ nonzerodivisors we have*

$$\partial_A(a,b) = \prod Norm_{\kappa(\mathfrak{m}_j)/\kappa(\mathfrak{m})}(\partial_{B_j}(a,b))$$

where the product is over the maximal ideals $\mathfrak{m}_j$ of B and $B_j = B_{\mathfrak{m}_j}$.

Proof. Choose $B_j \subset C_j$ as in Lemma 0EAG for a, b. By Lemma 0EAE we can choose a finite ring extension $B \subset C$ with $C_j \cong C_{\mathfrak{m}_j}$ for all j. Let $\mathfrak{m}_{j,k} \subset C$ be the maximal ideals of C lying over $\mathfrak{m}_j$. Let

$$a = u_{j,k}\pi_{j,k}^{f_{j,k}}, \quad b = v_{j,k}\pi_{j,k}^{g_{j,k}}$$

be the local factorizations which exist by our choice of $C_j \cong C_{\mathfrak{m}_j}$. By definition we have

$$\partial_A(a,b) = \prod\nolimits_{j,k} \mathrm{Norm}_{\kappa(\mathfrak{m}_{j,k})/\kappa(\mathfrak{m})}((-1)^{f_{j,k}g_{j,k}} u_{j,k}^{g_{j,k}} v_{j,k}^{-f_{j,k}} \bmod \mathfrak{m}_{j,k})^{m_{j,k}}$$

and

$$\partial_{B_j}(a,b) = \prod\nolimits_{k} \mathrm{Norm}_{\kappa(\mathfrak{m}_{j,k})/\kappa(\mathfrak{m}_j)}((-1)^{f_{j,k}g_{j,k}} u_{j,k}^{g_{j,k}} v_{j,k}^{-f_{j,k}} \bmod \mathfrak{m}_{j,k})^{m_{j,k}}$$

The result follows by transitivity of norms for $\kappa(\mathfrak{m}_{j,k})/\kappa(\mathfrak{m}_j)/\kappa(\mathfrak{m})$, see Fields, Lemma 0BIJ. □